Foundations of SHO Systems, Differential Equations, and Numerical Dynamics

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Mathematical Methods for Physicists II

Coursework Notes and Computational Guides

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This is a collection of notes from my Mathematical Methods for Physicists II coursework on oscillators, numerical integration, pendulums, and nonlinear dynamics. It provides the conceptual foundations for notebooks 2A and 2B, including their numerical extensions. These notes are a learning companion rather than a substitute for a textbook.

1 From random motion to deterministic dynamics

The probability, ensemble, and diffusion concepts developed for notebooks 1A and 1B remain useful background; they are not repeated here. In those notebooks a random draw selected each displacement. In 2A and 2B, the equations determine the acceleration from the present state. A time-stepping rule approximates their continuous evolution. No random number generator is required, even for the chaotic pendulum.

The central questions are: what is the state, which equation determines its rate of change, how do we advance it, and how do we check the approximation? Read this guide before the separate exercise guides. A numerical plot becomes physically meaningful only after these questions have answers.

2 Differential equations and the state

An ordinary differential equation (ODE) describes change with respect to one independent variable, here time. An initial-value problem specifies both an equation and enough initial data to select a solution. For Newtonian motion,

$$m\ddot{x} = F(t, x, \dot{x}), \qquad x(0) = x_0, \quad \dot{x}(0) = v_0.$$

Define velocity $v = \dot{x}$. The second-order equation becomes

$$\dot{x} = v, \qquad \dot{v} = a(t, x, v) = F/m.$$

The state is the ordered pair $\mathbf{y} = (x, v)$ and its derivative is $\mathbf{f} = (v, a)$. For a pendulum use $\mathbf{y} = (\theta, \omega)$, where $\omega = \dot{\theta}$. Position alone is insufficient: the same position with opposite velocities gives different future motion.

A numerical solver stores states at $t_n = nh$. The step size h is not the simulation duration. For a duration T with T/h integral, take $N = T/h$ steps and store $N + 1$ states, including the initial state. If T/h is not integral, rounding changes the endpoint; choose an exactly dividing step or handle a shorter final step explicitly.

3 Forces, equilibria, and potentials

An equilibrium has zero velocity and zero acceleration. A restoring force pushes toward equilibrium from either side. Hooke's law,

$$F = -k(x - x_{\text{eq}}),$$

has a stable equilibrium for $k > 0$. In one dimension, a conservative force can be written $F = -dU/dx$. In a per-unit-mass formulation write $a = -du/dx$, where $u = U/m$.

For the scaled nonlinear examples $\ddot{x} = -x^n$,

$$u(x) = \frac{x^{n+1}}{n+1}.$$

For $n = 1$, this is a quadratic well; for $n = 3$, a quartic well. For $n = 2$, $u = x^3/3$ decreases without bound as $x \to -\infty$. The force $-x^2$ is always nonpositive and does not restore negative positions toward zero. A runaway solution is therefore possible. These equations use scaled variables; a dimensional power-law model would need a coefficient with suitable units.

4 The harmonic oscillator and an exact benchmark

With $c = k/m > 0$, the equation is $\ddot{x} = -cx$ and its natural angular frequency is $\Omega_0 = \sqrt{c}$. The exact solution is

$$x(t) = x_0 \cos(\Omega_0 t) + \frac{v_0}{\Omega_0} \sin(\Omega_0 t),$$
$$v(t) = -x_0 \Omega_0 \sin(\Omega_0 t) + v_0 \cos(\Omega_0 t).$$

The period is $2\pi/\Omega_0$. For 2A's first examples, $c = 5, x_0 = 1, v_0 = 0$.

The per-unit-mass energy is $e = (v^2 + cx^2)/2$. Differentiating gives $\dot{e} = v(-cx) + cxv = 0$. For a spring in SI units, total energy is $E = mv^2/2 + k(x - x_{\mathrm{eq}})^2/2$. The mass factor distinguishes these two quantities. A constant numerical energy does not guarantee the correct phase or period, so compare trajectories as well.

5 Numerical accuracy and numerical stability

A Taylor expansion motivates time stepping:

$$\mathbf{y}(t+h) = \mathbf{y}(t) + h\dot{\mathbf{y}}(t) + \tfrac{1}{2}h^2\ddot{\mathbf{y}}(t) + \cdots .$$

A method of global order p usually has error proportional to h^p on a fixed finite interval in its asymptotic regime. Its one-step local error, starting from the exact state, is typically $O(h^{p+1})$. Accumulation and amplification of errors determine the global behaviour.

Method	Local error	Global order
Forward Euler	$O(h^2)$	1
Midpoint RK2	$O(h^3)$	2
Classical RK4	$O(h^5)$	4
Velocity Verlet	$O(h^3)$	2

Accuracy asks how close an approximation is; stability asks whether unwanted numerical perturbations grow. A smaller step usually helps, but a coarse unstable run need not display its formal convergence order. A high-order method also requires sufficiently small steps to resolve the fastest relevant motion.

6 Forward Euler: follow the present slope

For $\dot{\mathbf{y}} = \mathbf{f}(t, \mathbf{y})$,

$$\mathbf{y}_{n+1} = \mathbf{y}_n + h\mathbf{f}(t_n, \mathbf{y}_n).$$

For the oscillator,

$$x_{n+1} = x_n + hv_n, \qquad v_{n+1} = v_n - hcx_n.$$

Both slopes use the old state. Use temporary variables before committing the updates. Substituting these formulas into energy gives

$$e_{n+1} = (1 + ch^2)e_n.$$

Thus forward Euler artificially increases harmonic-oscillator energy at every nonzero step size. Smaller h slows the growth on a fixed interval; it does not make the method energy conserving.

7 Midpoint RK2: predict, then correct the slope

First estimate the state halfway through the step, then evaluate the slope there:

$$\mathbf{k}_1 = \mathbf{f}(t_n, \mathbf{y}_n), \quad \mathbf{k}_2 = \mathbf{f}(t_n + h/2, \mathbf{y}_n + h\mathbf{k}_1/2), \quad \mathbf{y}_{n+1} = \mathbf{y}_n + h\mathbf{k}_2.$$

For the oscillator the midpoint pair is $(x_n + hv_n/2,\ v_n - hcx_n/2)$. Both derivatives must use that pair. These $\mathbf{k}$ values are slopes, not increments; the factor h is supplied separately.

RK2 improves trajectory accuracy but is not an exact energy-conserving method. For the harmonic oscillator its energy multiplier is $1 + c^2h^4/4$, smaller than Euler's artificial gain for sufficiently small steps.

8 Classical RK4: four coupled estimates

The four stages are

$$\begin{aligned}
\mathbf{k}_1 &= \mathbf{f}(t_n, \mathbf{y}_n),\\
\mathbf{k}_2 &= \mathbf{f}(t_n + h/2, \mathbf{y}_n + h\mathbf{k}_1/2),\\
\mathbf{k}_3 &= \mathbf{f}(t_n + h/2, \mathbf{y}_n + h\mathbf{k}_2/2),\\
\mathbf{k}_4 &= \mathbf{f}(t_n + h, \mathbf{y}_n + h\mathbf{k}_3),\\
\mathbf{y}_{n+1} &= \mathbf{y}_n + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4).
\end{aligned}$$

The weights are selected to match the solution's Taylor expansion through fourth order. For nonlinear acceleration evaluate the force at each intermediate position, not repeatedly at x_n. If a force depends explicitly on time, also advance its time argument at each stage.

9 Velocity Verlet: position, force, then velocity

For acceleration depending on position,

$$\begin{aligned}
x_{n+1} &= x_n + hv_n + \tfrac{1}{2}h^2 a_n,\\
a_{n+1} &= a(x_{n+1}),\\
v_{n+1} &= v_n + \tfrac{h}{2}(a_n + a_{n+1}).
\end{aligned}$$

The new position allows the new force to be evaluated; averaging the two accelerations completes the velocity update. For a conservative position force, velocity Verlet is second order, time reversible, and symplectic: it preserves the phase-space geometry underlying Hamiltonian motion. This often yields bounded oscillatory energy error on stable long runs, rather than exact conservation.

For a harmonic oscillator, stable oscillatory integration requires $h\Omega_0 < 2$; practical accuracy requires a substantially smaller value. Standard velocity Verlet cannot simply be reused unchanged for velocity-dependent damping.

10 A pendulum from torque to acceleration

For a point mass on a rigid massless rod, gravitational torque is $-mgL\sin\theta$ and moment of inertia is mL^2. Therefore

$$\ddot{\theta} = -\frac{g}{L}\sin\theta.$$

Angles must be in radians. Near the downward equilibrium, $\sin\theta \approx \theta$, giving $\Omega_0 = \sqrt{g/L}$. At larger amplitudes the full sine matters and the libration period grows with amplitude.

Libration means back-and-forth motion; rotation means continuing through complete turns. Phase space (θ, ω) has a circular angle coordinate: points separated by 2π in angle represent the same orientation.

11 Pendulum energy and phase portraits

Taking potential zero at the bottom,

$$E_{\text{physical}} = \tfrac{1}{2}mL^2\omega^2 + mgL(1 - \cos\theta).$$

The notebooks calculate the reduced quantity

$$\mathcal{E} = \frac{E_{\text{physical}}}{mL^2} = \tfrac{1}{2}\omega^2 + \frac{g}{L}(1 - \cos\theta),$$

with units s^{-2}. It is not energy in joules, nor energy per unit mass. Multiply by mL^2 for joules or by L^2 for energy per unit mass.

Without damping or driving, $\mathcal{E}$ is constant. Closed phase curves correspond to librations. The upright stationary state has $\mathcal{E} = 2g/L$ and lies on the separatrix between libration and rotation. Numerical drift can distort these curves.

12 Damping and periodic driving

The model in 2B is

$$\dot{\theta} = \omega, \qquad \dot{\omega} = -\frac{g}{L}\sin\theta - q\omega + F_D \sin(\Omega_D t).$$

Here q has units s^{-1}, F_D is an angular acceleration with units s^{-2}, and Ω_D has units s^{-1}. The energy balance is

$$\dot{\mathcal{E}} = -q\omega^2 + F_D\omega \sin(\Omega_D t).$$

Damping removes energy; the drive can supply or remove it depending on the relative phase. Constant energy is not expected in a damped or driven run.

For the linearized free damped equation, roots obey $r^2 + qr + \Omega_0^2 = 0$. The motion is underdamped for $q < 2\Omega_0$, critically damped at equality, and overdamped above it. In strong overdamping the slow return rate is approximately Ω_0^2/q; increasing damping can slow the final return.

The linear steady-state amplitude under a sinusoidal drive is

$$A = \frac{F_D}{\sqrt{(\Omega_0^2 - \Omega_D^2)^2 + (q\Omega_D)^2}}.$$

This formula applies to the small-angle regime, not arbitrary nonlinear motion. A transient depends on the initial condition; the eventual response depends on the drive and damping, and in nonlinear systems can also depend on the attractor basin.

13 Euler–Cromer and angle representation

The main pendulum solver uses Euler–Cromer, also called semi-implicit Euler:

$$\begin{aligned} a_n &= -\frac{g}{L}\sin\theta_n - q\omega_n + F_D \sin(\Omega_D t_n), \\ \omega_{n+1} &= \omega_n + h a_n, \\ \theta_{n+1} &= \theta_n + h\omega_{n+1}. \end{aligned}$$

The acceleration uses old angle, velocity, and time, but the angle update uses the new velocity. This makes it a different method from forward Euler. It remains first order. For the undamped conservative problem it is symplectic; the damped update is dissipative.

Wrapping uses $\theta_{\mathrm{w}} = ((\theta + \pi) \bmod 2\pi) - \pi$. A jump at $\pm\pi$ is a coordinate discontinuity. An unwrapped angle records accumulated revolutions. The shortest orientation difference is

$$d_\theta = |((\theta_A - \theta_B + \pi) \bmod 2\pi) - \pi| \le \pi.$$

Neither a small d_θ nor a large unwrapped difference tells the whole story: velocity and rotation count matter too.

14 Deterministic chaos and nearby trajectories

Chaos means deterministic evolution with sensitive dependence on initial conditions in the relevant sustained dynamics. It does not mean random forcing, and an irregular plot alone is insufficient evidence. A small perturbation may initially grow approximately as $\delta(t) \sim \delta_0 e^{\lambda t}$, with positive largest Lyapunov exponent in a chaotic regime.

To compare driven trajectories, hold their driving phase and all parameters fixed. Their full extended state is (θ, ω, ϕ) with $\dot{\phi} = \Omega_D$ and ϕ taken modulo 2π. This explains why the periodic drive makes the system more complex than an autonomous two-variable damped pendulum.

Linearizing two nearby solutions at the same drive phase gives

$$\delta\dot{\theta} = \delta\omega, \qquad \delta\dot{\omega} = -\frac{g}{L}\cos\theta(t)\,\delta\theta - q\,\delta\omega.$$

The coefficient depends on the reference trajectory. Damping alone does not ensure every perturbation contracts; nonlinear driven motion can produce stretching in some directions.

A phase-space distance can be defined as

$$d = \sqrt{d_\theta^2 + (\delta\omega/\Omega_0)^2}.$$

Scaling velocity by a frequency puts both terms on comparable dimensionless angular scales. This is a possible diagnostic, not the quantity computed in the notebook's separation plots.

15 Step-size refinement: what it can establish

15.1 A discretization is another model

A fixed-step solver replaces a continuous flow by a discrete map. Large steps can change its apparent stability, energy balance, period, or attractor. Repeating at $h, h/2, h/4$ asks whether the reported behaviour belongs to the differential equation or to the chosen discretization. Keep the same initial state, parameters, physical duration, and comparisons.

For a regular solution on a fixed interval, if

$$y_h(t) = y(t) + C(t)h^p + \text{higher-order terms},$$

then successive differences approximately obey

$$\frac{|y_h - y_{h/2}|}{|y_{h/2} - y_{h/4}|} \approx 2^p.$$

The finer result has estimated leading error $(y_h - y_{h/2})/(2^p - 1)$ in magnitude. This argument needs the asymptotic regime and comparable times; it is unreliable after strong chaotic amplification or near cancellation. It is conceptual background, not an extrapolation implemented in 2B.

15.2 Why chaotic trajectories eventually stop agreeing

Truncation and roundoff act as small perturbations. If their effect grows like $e^{\lambda t}$, an initial error ϵ reaches tolerance Δ after roughly

$$t_{\text{predict}} \sim \lambda^{-1}\log(\Delta/\epsilon).$$

Reducing h extends this useful prediction interval but need not produce matching trajectories indefinitely. Limits at fixed time and over arbitrarily long time should not be confused.

Test early-time agreement and the persistence of qualitative features or sufficiently sampled statistical summaries. Similar attractor pictures across steps are supportive, not a convergence proof. Near a bifurcation, discretization can also move a trajectory between attractor basins. Comparing an independent accurate solver would provide a stronger further check.

15.3 Reading semilogarithmic separation plots

On a logarithmic vertical axis, exponential growth can appear as an approximately straight rising segment. However, 2B plots orientation separation only, which can shrink when angles align even if velocities differ. It also saturates at π. The plot floor 10^{-15} merely allows zero values to be displayed; it is not a measured nonzero separation. Avoid fitting saturated, transient, or floor-limited intervals.

A reliable largest-Lyapunov estimate would evolve the full perturbation, repeatedly renormalize it, and average logarithmic growth after transients. The notebook does not implement that procedure; its plots are a sensitivity demonstration.

16 Poincaré sections: observe at the same drive phase

16.1 The stroboscope idea

A full phase portrait overlays motion during all drive phases. Instead imagine taking a photograph once every forcing period,

$$T_D = \frac{2\pi}{\Omega_D}.$$

Sampling at $t_n = t_0 + nT_D$ fixes $\phi = \Omega_D t_0 \bmod 2\pi$. The sampled pairs define the stroboscopic Poincaré map

$$(\theta_{n+1}, \omega_{n+1}) = P(\theta_n, \omega_n).$$

This is a section of the extended state at fixed drive phase. It is not a histogram and not an arbitrary downsampling of a trajectory.

16.2 What different patterns mean

A drive-periodic response is a fixed point of P: each photograph repeats the same state. A response of period rT_D gives an r-cycle, hence r repeating points. This statement uses angle modulo 2π; a running solution can repeat its orientation and velocity while advancing whole revolutions.

A quasiperiodic response can produce a smooth invariant curve, and chaotic motion can produce a structured distributed set. A finite cloud also admits other explanations: residual transient, noise, a poorly resolved solver, or a high-period orbit. A picture alone cannot determine a Lyapunov exponent or establish fractal dimension.

16.3 Transients, attraction, and dissipation

An attractor is a set approached by trajectories from a region of initial conditions called its basin. Damping helps discard memory of some initial conditions, but nonlinear driven systems may have multiple attractors. In the (θ, ω) dynamics, the divergence is $-q$, so an infinitesimal phase-space area contracts by e^{-qT_D} over one period of the exact flow. Simultaneous stretching and contraction can organize a complex dissipative attractor.

Discard an initial number of periods before plotting. There is no universal discard count: extend the transient and test whether the retained pattern changes. The notebook uses 100 discarded and 300 retained periods as a practical exercise choice, not a guarantee.

16.4 Sampling without phase drift

Choose an integer M steps per period and set $h = T_D/M$. The index of the nth sample is exactly nM when the initial state has index zero. This avoids a slowly drifting phase caused by choosing every rounded T_D/h steps with an incompatible h. If an arbitrary time grid is necessary, interpolate to the target times with appropriate accuracy.

For $\Omega_D = 2/3$, $T_D = 3\pi \approx 9.42478$. The notebook uses $M = 240, 480, 960$, giving step sizes approximately 0.03927, 0.01963, and 0.00982. Its retained indices are $101M, 102M, \ldots, 400M$, exactly 300 points. The first 100 periods are omitted. Integrate the angle unwrapped, then wrap only the sampled positions.

17 Designing a defensible numerical experiment

For an exact benchmark, compare maximum trajectory error, endpoint error, energy behaviour, and refinement at a fixed duration. In 2A define

$$\epsilon_\infty(h) = \max_n |x_h(t_n) - x_{\text{exact}}(t_n)|, \quad p_{\text{obs}} = \log_2 \frac{\epsilon_\infty(h)}{\epsilon_\infty(h/2)}.$$

A small endpoint error can result from cancellation of amplitude and phase errors. A full-interval check is more informative.

For a nonlinear driven problem without a simple exact solution, use successive step sizes, physically appropriate diagnostics, and consistent sampling. Preserve the original scientific question when comparing methods; distinguish numerical energy drift from physical damping or drive work. Before coding, write the state, its derivatives, the update order, the output times, and the intended checks on paper.

18 Book references and reading route

1. Nicholas J. Giordano and Hisao Nakanishi, *Computational Physics*, 2nd edition, Addison-Wesley (2005), Chapter 3. This is the coursework reference for oscillatory motion and the driven pendulum. The figure numbers in 2B retain the original homework labels; they are not a claim of exact reproduction.
2. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. Read the discussions of phase planes, limit cycles, Poincaré maps, and chaos for the dynamical interpretation of 2B. Topic titles are given because page and chapter locations vary by edition.
3. William H. Press, Saul A. Teukolsky, William T. Vetterling, and Brian P. Flannery, *Numerical Recipes: The Art of Scientific Computing*, 3rd edition, Cambridge University Press (2007), Chapter 17. Sections 17.1–17.4 cover Runge–Kutta, step-size control, extrapolation, and conservative second-order equations. Official contents: https://numerical.recipes/book/.

These notes are an independent explanation of the coursework, not extracts from these books. Begin with Giordano and Nakanishi for computational examples, use Strogatz for conceptual dynamics, and consult Numerical Recipes for numerical error control.